Airbnb Data Analysis

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```
#install.packages(c("tidyverse", "caret", "glmnet", "MASS", "car", "class"))
#install.packages("LaTeX")

library(tidyverse)

## Warning: package 'tidyverse' was built under R version 4.4.3

## Warning: package 'ggplot2' was built under R version 4.4.3

## Warning: package 'tidyr' was built under R version 4.4.3

## Warning: package 'readr' was built under R version 4.4.3

## Warning: package 'purrr' was built under R version 4.4.3

## Warning: package 'forcats' was built under R version 4.4.3

## Warning: package 'lubridate' was built under R version 4.4.3

## ── Attaching core tidyverse packages ──────────────────────── tidyverse
2.0.0 ──
## ✓ dplyr     1.1.4     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.2     ✓ tibble    3.2.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.1
## ✓ purrr     1.2.1
## ── Conflicts ────────────────────────────────────────
tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

library(caret)

## Warning: package 'caret' was built under R version 4.4.3

## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```r
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.4.3
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Loaded glmnet 4.1-10
```

```r
library(MASS)
```

```
## Warning: package 'MASS' was built under R version 4.4.3
```

```
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
```

```r
library(car)
```

```
## Warning: package 'car' was built under R version 4.4.3
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.4.3
```

```
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
##
##     some
```

```r
df <- read.csv("final_data_with_changes.csv")

head(df)
```

```
##   price_total    room_type is_shared_room is_private_room max_guests
## 1    93.78818 Private room              0               1          2
## 2   822.16905 Private room              0               1          4
## 3   104.35183 Private room              0               1          2
```

```
## 4  1031.98604 Private room             0              1          4
## 5  1078.98356 Private room             0              1          2
## 6   463.15305 Private room             0              1          3
##   is_superhost is_multi_listing is_business_listing cleanliness_score
## 1            1                1                   0                10
## 2            1                0                   0                 8
## 3            1                0                   1                 9
## 4            1                0                   1                 9
## 5            0                0                   0                10
## 6            1                0                   0                 8
##   guest_satisfaction_score num_bedrooms distance_city_center
distance_metro
## 1                       93            1            5.0229638
2.5393800
## 2                       85            1            0.4883893
0.2394039
## 3                       87            1            5.7483119
3.6516213
## 4                       90            2            0.3848620
0.4398761
## 5                       98            1            0.5447382
0.3186926
## 6                      100            2            2.1314201
1.9046682
##   attraction_index attraction_index_norm restaurant_index
restaurant_index_norm
## 1         78.69038              4.166708         98.25390
6.846473
## 2        631.17638             33.421209        837.28076
58.342928
## 3         75.27588              3.985908         95.38695
6.646700
## 4        493.27253             26.119108        875.03310
60.973565
## 5        552.83032             29.272733        815.30574
56.811677
## 6        174.78896              9.255191        225.20166
15.692376
##   longitude latitude      city day_type           district         state
## 1   4.90569 52.41772 Amsterdam  weekday Gemeente Landsmeer North Holland
## 2   4.90005 52.37432 Amsterdam  weekday Gemeente Amsterdam North Holland
## 3   4.97512 52.36103 Amsterdam  weekday    Gemeente Diemen North Holland
## 4   4.89417 52.37663 Amsterdam  weekday Gemeente Amsterdam North Holland
## 5   4.90051 52.37508 Amsterdam  weekday Gemeente Amsterdam North Holland
## 6   4.87699 52.38966 Amsterdam  weekday Gemeente Amsterdam North Holland
##   country_code country_name Crime_Index Safety_Index
Monthly_Average_Net_salary
## 1           NL  Netherlands        25.7         74.3
4502.1
## 2           NL  Netherlands        25.7         74.3
```

```
4502.1
## 3           NL  Netherlands        25.7         74.3
4502.1
## 4           NL  Netherlands        25.7         74.3
4502.1
## 5           NL  Netherlands        25.7         74.3
4502.1
## 6           NL  Netherlands        25.7         74.3
4502.1
##   Meal_at_Inexpensive_Restaurant Taxi_price_per_Km Monthly_Basic_Utilities
## 1                           20.5                3                      262
## 2                           20.5                3                      262
## 3                           20.5                3                      262
## 4                           20.5                3                      262
## 5                           20.5                3                      262
## 6                           20.5                3                      262
##   Monthly_Rent_One_Bedroom_CC Monthly_Rent_One_Bedroom_OCC
## 1                     2178.68                      1701.26
## 2                     2178.68                      1701.26
## 3                     2178.68                      1701.26
## 4                     2178.68                      1701.26
## 5                     2178.68                      1701.26
## 6                     2178.68                      1701.26
##   Monthly_Rent_Three_Bedroom_CC Monthly_Rent_Three_Bedroom_OCC
proximity_index
## 1                        3725.5                        2785.29
0.2254479
## 2                        3725.5                        2785.29
1.1139552
## 3                        3725.5                        2785.29
0.1842912
## 4                        3725.5                        2785.29
1.1102748
## 5                        3725.5                        2785.29
1.0364622
## 6                        3725.5                        2785.29
0.3907734
```

```
str(df)
```

```
## 'data.frame':    51707 obs. of  36 variables:
##  $ price_total                  : num  93.8 822.2 104.4 1032 1079 ...
##  $ room_type                    : chr  "Private room" "Private room"
"Private room" "Private room" ...
##  $ is_shared_room               : int  0 0 0 0 0 0 0 0 0 0 ...
##  $ is_private_room              : int  1 1 1 1 1 1 1 0 0 1 ...
##  $ max_guests                   : num  2 4 2 4 2 3 2 4 4 2 ...
##  $ is_superhost                 : int  1 1 1 1 0 1 1 0 1 1 ...
##  $ is_multi_listing             : int  1 0 0 0 0 0 0 0 0 1 ...
##  $ is_business_listing          : int  0 0 1 1 0 0 0 0 0 0 ...
```

```
##  $ cleanliness_score             : num  10 8 9 9 10 8 10 10 9 10 ...
##  $ guest_satisfaction_score      : num  93 85 87 90 98 100 94 100 96 88
...
##  $ num_bedrooms                  : int  1 1 1 2 1 2 1 3 2 1 ...
##  $ distance_city_center          : num  5.023 0.488 5.748 0.385 0.545 ...
##  $ distance_metro                : num  2.539 0.239 3.652 0.44 0.319 ...
##  $ attraction_index              : num  78.7 631.2 75.3 493.3 552.8 ...
##  $ attraction_index_norm         : num  4.17 33.42 3.99 26.12 29.27 ...
##  $ restaurant_index              : num  98.3 837.3 95.4 875 815.3 ...
##  $ restaurant_index_norm         : num  6.85 58.34 6.65 60.97 56.81 ...
##  $ longitude                     : num  4.91 4.9 4.98 4.89 4.9 ...
##  $ latitude                      : num  52.4 52.4 52.4 52.4 52.4 ...
##  $ city                          : chr  "Amsterdam" "Amsterdam"
"Amsterdam" "Amsterdam" ...
##  $ day_type                      : chr  "weekday" "weekday" "weekday"
"weekday" ...
##  $ district                      : chr  "Gemeente Landsmeer" "Gemeente
Amsterdam" "Gemeente Diemen" "Gemeente Amsterdam" ...
##  $ state                         : chr  "North Holland" "North Holland"
"North Holland" "North Holland" ...
##  $ country_code                  : chr  "NL" "NL" "NL" "NL" ...
##  $ country_name                  : chr  "Netherlands" "Netherlands"
"Netherlands" "Netherlands" ...
##  $ Crime_Index                   : num  25.7 25.7 25.7 25.7 25.7 25.7 25.7
25.7 25.7 25.7 ...
##  $ Safety_Index                  : num  74.3 74.3 74.3 74.3 74.3 74.3 74.3
74.3 74.3 74.3 ...
##  $ Monthly_Average_Net_salary    : num  4502 4502 4502 4502 4502 ...
##  $ Meal_at_Inexpensive_Restaurant: num  20.5 20.5 20.5 20.5 20.5 20.5 20.5
20.5 20.5 20.5 ...
##  $ Taxi_price_per_Km             : num  3 3 3 3 3 3 3 3 3 3 ...
##  $ Monthly_Basic_Utilities       : num  262 262 262 262 262 262 262 262
262 262 ...
##  $ Monthly_Rent_One_Bedroom_CC   : num  2179 2179 2179 2179 2179 ...
##  $ Monthly_Rent_One_Bedroom_OCC  : num  1701 1701 1701 1701 1701 ...
##  $ Monthly_Rent_Three_Bedroom_CC : num  3726 3726 3726 3726 3726 ...
##  $ Monthly_Rent_Three_Bedroom_OCC: num  2785 2785 2785 2785 2785 ...
##  $ proximity_index               : num  0.225 1.114 0.184 1.11 1.036 ...
```

```
summary(df)
```

```
##   price_total        room_type         is_shared_room     is_private_room
##  Min.   :    7.66   Length:51707       Min.   :0.000000   Min.   :0.0000
##  1st Qu.:  157.54   Class :character   1st Qu.:0.000000   1st Qu.:0.0000
##  Median :  260.50   Mode  :character   Median :0.000000   Median :0.0000
##  Mean   :  368.62                      Mean   :0.007078   Mean   :0.3615
##  3rd Qu.:  433.67                      3rd Qu.:0.000000   3rd Qu.:1.0000
##  Max.   :43955.62                      Max.   :1.000000   Max.   :1.0000
##    max_guests     is_superhost    is_multi_listing is_business_listing
##  Min.   :2.000   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
```

```
##  1st Qu.:2.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
##  Median :3.000   Median :1.0000   Median :0.0000   Median :0.0000
##  Mean   :3.162   Mean   :0.7441   Mean   :0.2914   Mean   :0.3502
##  3rd Qu.:4.000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000
##  Max.   :6.000   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
##  cleanliness_score guest_satisfaction_score  num_bedrooms
##  Min.   : 2.000    Min.   : 20.00           Min.   : 0.000
##  1st Qu.: 9.000    1st Qu.: 90.00           1st Qu.: 1.000
##  Median :10.000    Median : 95.00           Median : 1.000
##  Mean   : 9.391    Mean   : 92.63           Mean   : 1.159
##  3rd Qu.:10.000    3rd Qu.: 99.00           3rd Qu.: 1.000
##  Max.   :10.000    Max.   :100.00           Max.   :10.000
##  distance_city_center distance_metro      attraction_index
##  Min.   : 0.01504     Min.   : 0.002301   Min.   :  15.15
##  1st Qu.: 1.45314     1st Qu.: 0.248480   1st Qu.: 136.80
##  Median : 2.61354     Median : 0.413269   Median : 234.33
##  Mean   : 3.19129     Mean   : 0.681540   Mean   : 294.20
##  3rd Qu.: 4.26308     3rd Qu.: 0.737840   3rd Qu.: 385.76
##  Max.   :25.28456     Max.   :14.273577   Max.   :4513.56
##  attraction_index_norm restaurant_index  restaurant_index_norm
##  Min.   :  0.9263      Min.   :  19.58   Min.   :  0.5928
##  1st Qu.:  6.3809      1st Qu.: 250.85   1st Qu.:  8.7515
##  Median : 11.4683      Median : 522.05   Median : 17.5422
##  Mean   : 13.4238      Mean   : 626.86   Mean   : 22.7862
##  3rd Qu.: 17.4151      3rd Qu.: 832.63   3rd Qu.: 32.9646
##  Max.   :100.0000      Max.   :6696.16   Max.   :100.0000
##    longitude          latitude         city              day_type
##  Min.   :-9.2263   Min.   :37.95   Length:51707       Length:51707
##  1st Qu.:-0.0725   1st Qu.:41.40   Class :character   Class :character
##  Median : 4.8730   Median :47.51   Mode  :character   Mode  :character
##  Mean   : 7.4261   Mean   :45.67
##  3rd Qu.:13.5188   3rd Qu.:51.47
##  Max.   :23.7860   Max.   :52.64
##    district            state           country_code       country_name
##  Length:51707       Length:51707       Length:51707       Length:51707
##  Class :character   Class :character   Class :character   Class :character
##  Mode  :character   Mode  :character   Mode  :character   Mode  :character
##
##
##
##   Crime_Index     Safety_Index   Monthly_Average_Net_salary
##  Min.   :25.70   Min.   :42.00   Min.   :1164
##  1st Qu.:33.90   1st Qu.:44.40   1st Qu.:1411
##  Median :47.30   Median :52.70   Median :2103
##  Mean   :46.37   Mean   :53.63   Mean   :2612
##  3rd Qu.:55.60   3rd Qu.:66.10   3rd Qu.:3305
##  Max.   :58.00   Max.   :74.30   Max.   :4502
##  Meal_at_Inexpensive_Restaurant Taxi_price_per_Km Monthly_Basic_Utilities
##  Min.   :10.47                  Min.   :0.900     Min.   :134.9
##  1st Qu.:15.00                  1st Qu.:1.150     1st Qu.:184.6
```

```
##  Median :15.00                  Median :1.500     Median :187.4
##  Mean   :16.50                  Mean   :1.781     Mean   :228.5
##  3rd Qu.:18.25                  3rd Qu.:2.800     3rd Qu.:291.2
##  Max.   :23.01                  Max.   :3.000     Max.   :347.1
##  Monthly_Rent_One_Bedroom_CC Monthly_Rent_One_Bedroom_OCC
##  Min.   : 614.4              Min.   : 519.2
##  1st Qu.:1280.5              1st Qu.: 820.6
##  Median :1387.5              Median :1036.0
##  Mean   :1518.0              Mean   :1105.0
##  3rd Qu.:1398.6              3rd Qu.:1061.2
##  Max.   :2788.4              Max.   :2029.8
##  Monthly_Rent_Three_Bedroom_CC Monthly_Rent_Three_Bedroom_OCC
proximity_index
##  Min.   :1143                  Min.   : 946.4                 Min.
:0.04745
##  1st Qu.:2163                  1st Qu.:1541.7                 1st
Qu.:0.43276
##  Median :2432                  Median :1633.3                 Median
:0.57557
##  Mean   :2693                  Mean   :1945.5                 Mean
:0.59753
##  3rd Qu.:3088                  3rd Qu.:2145.0                 3rd
Qu.:0.72529
##  Max.   :4403                  Max.   :3416.4                 Max.
:1.82096
```

```
names(df)
```

```
##  [1] "price_total"                    "room_type"
##  [3] "is_shared_room"                 "is_private_room"
##  [5] "max_guests"                     "is_superhost"
##  [7] "is_multi_listing"               "is_business_listing"
##  [9] "cleanliness_score"              "guest_satisfaction_score"
## [11] "num_bedrooms"                   "distance_city_center"
## [13] "distance_metro"                 "attraction_index"
## [15] "attraction_index_norm"          "restaurant_index"
## [17] "restaurant_index_norm"          "longitude"
## [19] "latitude"                       "city"
## [21] "day_type"                       "district"
## [23] "state"                          "country_code"
## [25] "country_name"                   "Crime_Index"
## [27] "Safety_Index"                   "Monthly_Average_Net_salary"
## [29] "Meal_at_Inexpensive_Restaurant" "Taxi_price_per_Km"
## [31] "Monthly_Basic_Utilities"        "Monthly_Rent_One_Bedroom_CC"
## [33] "Monthly_Rent_One_Bedroom_OCC"   "Monthly_Rent_Three_Bedroom_CC"
## [35] "Monthly_Rent_Three_Bedroom_OCC" "proximity_index"
```

```
colSums(is.na(df))
```

```
##                   price_total                     room_type
##                             0                             0
```

```
##                  is_shared_room                 is_private_room
##                               0                               0
##                     max_guests                    is_superhost
##                               0                               0
##                is_multi_listing             is_business_listing
##                               0                               0
##               cleanliness_score       guest_satisfaction_score
##                               0                               0
##                    num_bedrooms           distance_city_center
##                               0                               0
##                  distance_metro               attraction_index
##                               0                               0
##           attraction_index_norm               restaurant_index
##                               0                               0
##           restaurant_index_norm                      longitude
##                               0                               0
##                        latitude                           city
##                               0                               0
##                        day_type                       district
##                               0                               0
##                           state                   country_code
##                               0                               0
##                    country_name                    Crime_Index
##                               0                               0
##                    Safety_Index     Monthly_Average_Net_salary
##                               0                               0
## Meal_at_Inexpensive_Restaurant              Taxi_price_per_Km
##                               0                               0
##         Monthly_Basic_Utilities    Monthly_Rent_One_Bedroom_CC
##                               0                               0
##    Monthly_Rent_One_Bedroom_OCC  Monthly_Rent_Three_Bedroom_CC
##                               0                               0
## Monthly_Rent_Three_Bedroom_OCC                proximity_index
##                               0                               0

df <- na.omit(df)

df$log_price <- log(df$price)
```

The response variable (price) exhibited strong right-skewness. A logarithmic transformation was applied to normalize the distribution, stabilize variance, and improve the performance and interpretability of regression models.

```
#converting categorical variables
df <- df %>%
  mutate_if(is.character, as.factor)
```

All character variables were converted into factors to ensure proper handling of categorical predictors. This enables the regression model to internally generate dummy variables and incorporate qualitative features effectively.

```r
set.seed(123)

trainIndex <- createDataPartition(df$log_price, p = 0.7, list = FALSE)

train <- df[trainIndex, ]
test  <- df[-trainIndex, ]
```

The dataset was divided into training (70%) and testing (30%) subsets using stratified sampling. The training set was used to build predictive models, while the test set was used to evaluate model performance on unseen data.

```r
train_clean <- train %>% dplyr::select(-price_total)
test_clean  <- test  %>% dplyr::select(-price_total)
```

PHASE 3: MULTIPLE LINEAR REGRESSION

Y = $beta$0 + ...

fit model

```r
lm_model <- lm(log_price ~ ., data = train)

pred_lm <- predict(lm_model, newdata = test)

rmse <- function(actual, predicted) {
  sqrt(mean((actual - predicted)^2))
}

mae <- function(actual, predicted) {
  mean(abs(actual - predicted))
}

cat("Linear Regression:\n")

## Linear Regression:

cat("RMSE:", rmse(test$log_price, pred_lm), "\n")

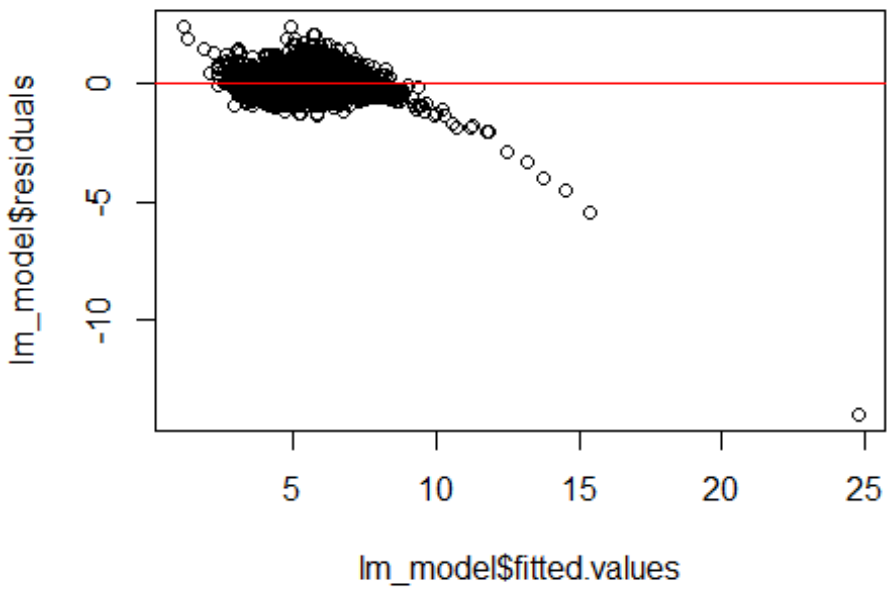
## RMSE: 0.2975413

cat("MAE:", mae(test$log_price, pred_lm), "\n")

## MAE: 0.2243146
```

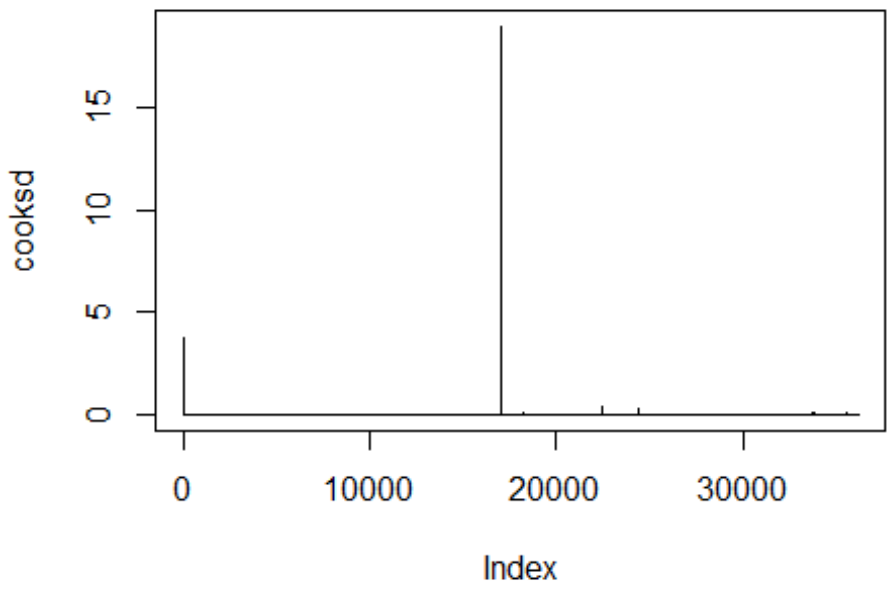
Residual Plot

```r
plot(lm_model$fitted.values, lm_model$residuals)
abline(h = 0, col = "red")
```



Cook's Distance

```
cooksd <- cooks.distance(lm_model)
plot(cooksd, type="h")
```



Stepwise Selection

```
step_model <- stepAIC(lm_model, direction = "both")

## Start:  AIC=-86449.1
## log_price ~ price_total + room_type + is_shared_room + is_private_room +
##     max_guests + is_superhost + is_multi_listing + is_business_listing +
##     cleanliness_score + guest_satisfaction_score + num_bedrooms +
##     distance_city_center + distance_metro + attraction_index +
##     attraction_index_norm + restaurant_index + restaurant_index_norm +
##     longitude + latitude + city + day_type + district + state +
##     country_code + country_name + Crime_Index + Safety_Index +
##     Monthly_Average_Net_salary + Meal_at_Inexpensive_Restaurant +
##     Taxi_price_per_Km + Monthly_Basic_Utilities +
Monthly_Rent_One_Bedroom_CC +
##     Monthly_Rent_One_Bedroom_OCC + Monthly_Rent_Three_Bedroom_CC +
##     Monthly_Rent_Three_Bedroom_OCC + proximity_index

## Warning in stepAIC(lm_model, direction = "both"): 0 df terms are changing
AIC

##
## Step:  AIC=-82111.09
## log_price ~ price_total + room_type + is_shared_room + is_private_room +
##     max_guests + is_superhost + is_multi_listing + is_business_listing +
##     cleanliness_score + guest_satisfaction_score + num_bedrooms +
##     distance_city_center + distance_metro + attraction_index +
##     attraction_index_norm + restaurant_index + restaurant_index_norm +
```

```
##     longitude + latitude + city + day_type + district + state +
##     country_code + country_name + Crime_Index + Safety_Index +
##     Monthly_Average_Net_salary + Meal_at_Inexpensive_Restaurant +
##     Taxi_price_per_Km + Monthly_Basic_Utilities +
Monthly_Rent_One_Bedroom_CC +
##     Monthly_Rent_One_Bedroom_OCC + Monthly_Rent_Three_Bedroom_CC +
##     Monthly_Rent_Three_Bedroom_OCC
```

```
summary(step_model)
```

```
##
## Call:
## lm(formula = log_price ~ price_total + room_type + is_shared_room +
##     is_private_room + max_guests + is_superhost + is_multi_listing +
##     is_business_listing + cleanliness_score + guest_satisfaction_score +
##     num_bedrooms + distance_city_center + distance_metro +
attraction_index +
##     attraction_index_norm + restaurant_index + restaurant_index_norm +
##     longitude + latitude + city + day_type + district + state +
##     country_code + country_name + Crime_Index + Safety_Index +
##     Monthly_Average_Net_salary + Meal_at_Inexpensive_Restaurant +
##     Taxi_price_per_Km + Monthly_Basic_Utilities +
Monthly_Rent_One_Bedroom_CC +
##     Monthly_Rent_One_Bedroom_OCC + Monthly_Rent_Three_Bedroom_CC +
##     Monthly_Rent_Three_Bedroom_OCC, data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.7458  -0.1970  -0.0139   0.1760   3.4732
##
## Coefficients: (50 not defined because of singularities)
##                                            Estimate Std. Error t value
## (Intercept)                                7.538e+11  1.314e+12   0.574
## price_total                                4.425e-04  3.842e-06 115.168
## room_typePrivate room                     -3.520e-01  4.640e-03 -75.867
## room_typeShared room                      -6.811e-01  2.025e-02 -33.636
## is_shared_room                                    NA         NA      NA
## is_private_room                                   NA         NA      NA
## max_guests                                 7.715e-02  1.886e-03  40.904
## is_superhost                              -1.822e-02  4.187e-03  -4.352
## is_multi_listing                           4.616e-02  4.356e-03  10.598
## is_business_listing                        1.096e-01  4.449e-03  24.637
## cleanliness_score                          3.654e-02  2.560e-03  14.271
## guest_satisfaction_score                   1.061e-04  2.790e-04   0.380
## num_bedrooms                               9.223e-02  3.418e-03  26.986
## distance_city_center                      -8.068e-02  1.584e-03 -50.947
## distance_metro                            -2.590e-01  3.323e-03 -77.938
## attraction_index                           3.012e-04  2.396e-05  12.572
## attraction_index_norm                      1.344e-03  5.557e-04   2.418
## restaurant_index                           5.424e-05  1.066e-05   5.091
```

```
## restaurant_index_norm                          6.639e-03  2.514e-04  26.412
## longitude                                     -5.696e-01  4.860e-02 -11.720
## latitude                                      -6.164e-01  8.576e-02  -7.187
## cityAthens                                     2.106e+11  3.367e+11   0.626
## cityBarcelona                                  2.018e+11  3.521e+11   0.573
## cityBerlin                                     7.613e+10  4.081e+11   0.187
## cityBudapest                                   1.042e+11  3.801e+11   0.274
## cityLisbon                                     9.506e+10  3.734e+11   0.255
## cityLondon                                     1.488e+11  4.814e+11   0.309
## cityParis                                      2.042e+11  3.521e+11   0.580
## cityRome                                       1.620e+11  2.719e+11   0.596
## cityVienna                                     3.597e+09  1.091e+11   0.033
## day_typeweekend                                3.267e-02  3.401e-03   9.606
## districtBerlin                                        NA         NA      NA
## districtBudapest                                      NA         NA      NA
## districtCitta metropolitana di Roma Capitale   2.209e-01  9.497e-03  23.254
## districtDepartement de Seine-Saint-Denis      -2.719e-02  1.562e-02  -1.741
## districtDepartement des Hauts-de-Seine         1.168e-01  1.205e-02   9.690
## districtDepartement du Val-de-Marne            4.534e-02  1.966e-02   2.306
## districtGemeente Aalsmeer                      6.501e-02  2.344e-01   0.277
## districtGemeente Amstelveen                   -5.562e-02  6.974e-02  -0.797
## districtGemeente Amsterdam                     7.095e-02  5.808e-02   1.222
## districtGemeente Diemen                        1.669e-01  6.732e-02   2.480
## districtGemeente Landsmeer                    -1.721e-02  7.319e-02  -0.235
## districtGemeente Ouder-Amstel                  1.150e-01  6.610e-02   1.740
## districtGemeente Waterland                     5.481e-02  1.548e-01   0.354
## districtGemeente Zaanstad                             NA         NA      NA
## districtGreater London                                NA         NA      NA
## districtLisbon                                 2.943e-02  1.901e-02   1.548
## districtLoures                                 1.202e-01  3.195e-02   3.761
## districtNomarchia Athinas                             NA         NA      NA
## districtOdivelas                               5.173e-02  4.233e-02   1.222
## districtOeiras                                 2.137e-01  3.544e-02   6.029
## districtParis                                         NA         NA      NA
## districtPolitischer Bezirk Ganserndorf        -1.821e-01  5.851e-02  -3.113
## districtPolitischer Bezirk Korneuburg         -4.524e-04  9.338e-02  -0.005
## districtPolitischer Bezirk Modling            -1.988e-01  3.390e-02  -5.866
## districtPolitischer Bezirk Wien-Umgebung      -6.140e-02  3.793e-02  -1.619
## districtProvincia de Barcelona                        NA         NA      NA
## districtRome                                          NA         NA      NA
## districtWien Stadt                                    NA         NA      NA
## stateBerlin                                    3.066e-02  1.446e-01   0.212
## stateBrandenburg                                      NA         NA      NA
## stateBudapest                                 -7.536e-01  8.008e-02  -9.411
## stateCatalonia                                        NA         NA      NA
## stateEngland                                          NA         NA      NA
## stateIle-de-France                                    NA         NA      NA
## stateLatium                                           NA         NA      NA
## stateLisbon                                           NA         NA      NA
## stateLower Austria                                    NA         NA      NA
```

```
## stateNorth Holland                                    NA          NA      NA
## statePest                                             NA          NA      NA
## stateRome                                             NA          NA      NA
## stateSetubal                                          NA          NA      NA
## stateVienna                                           NA          NA      NA
## country_codeDE                                        NA          NA      NA
## country_codeES                                        NA          NA      NA
## country_codeFR                                        NA          NA      NA
## country_codeGB                                        NA          NA      NA
## country_codeGR                                        NA          NA      NA
## country_codeHU                                        NA          NA      NA
## country_codeIT                                        NA          NA      NA
## country_codeNL                                        NA          NA      NA
## country_codePT                                        NA          NA      NA
## country_codeVA                                        NA          NA      NA
## country_nameFrance                                    NA          NA      NA
## country_nameGermany                                   NA          NA      NA
## country_nameGreece                                    NA          NA      NA
## country_nameHoly See (Vatican City State)             NA          NA      NA
## country_nameHungary                                   NA          NA      NA
## country_nameItaly                                     NA          NA      NA
## country_nameNetherlands                               NA          NA      NA
## country_namePortugal                                  NA          NA      NA
## country_nameSpain                                     NA          NA      NA
## country_nameUnited Kingdom                            NA          NA      NA
## Crime_Index                                    -1.313e+10  2.258e+10  -0.582
## Safety_Index                                   -7.127e+09  1.858e+10  -0.384
## Monthly_Average_Net_salary                            NA          NA      NA
## Meal_at_Inexpensive_Restaurant                        NA          NA      NA
## Taxi_price_per_Km                                     NA          NA      NA
## Monthly_Basic_Utilities                        4.321e+08  3.196e+09   0.135
## Monthly_Rent_One_Bedroom_CC                           NA          NA      NA
## Monthly_Rent_One_Bedroom_OCC                          NA          NA      NA
## Monthly_Rent_Three_Bedroom_CC                         NA          NA      NA
## Monthly_Rent_Three_Bedroom_OCC                        NA          NA      NA
##                                               Pr(>|t|)
## (Intercept)                                    0.56630
## price_total                                   < 2e-16 ***
## room_typePrivate room                         < 2e-16 ***
## room_typeShared room                          < 2e-16 ***
## is_shared_room                                     NA
## is_private_room                                    NA
## max_guests                                    < 2e-16 ***
## is_superhost                                  1.36e-05 ***
## is_multi_listing                              < 2e-16 ***
## is_business_listing                           < 2e-16 ***
## cleanliness_score                             < 2e-16 ***
## guest_satisfaction_score                       0.70381
## num_bedrooms                                  < 2e-16 ***
## distance_city_center                          < 2e-16 ***
```

```
## distance_metro                                  < 2e-16 ***
## attraction_index                                < 2e-16 ***
## attraction_index_norm                           0.01561 *
## restaurant_index                               3.58e-07 ***
## restaurant_index_norm                           < 2e-16 ***
## longitude                                       < 2e-16 ***
## latitude                                       6.74e-13 ***
## cityAthens                                      0.53164
## cityBarcelona                                   0.56658
## cityBerlin                                      0.85201
## cityBudapest                                    0.78408
## cityLisbon                                      0.79905
## cityLondon                                      0.75727
## cityParis                                       0.56187
## cityRome                                        0.55131
## cityVienna                                      0.97371
## day_typeweekend                                 < 2e-16 ***
## districtBerlin                                       NA
## districtBudapest                                     NA
## districtCitta metropolitana di Roma Capitale    < 2e-16 ***
## districtDepartement de Seine-Saint-Denis        0.08168 .
## districtDepartement des Hauts-de-Seine          < 2e-16 ***
## districtDepartement du Val-de-Marne             0.02111 *
## districtGemeente Aalsmeer                       0.78148
## districtGemeente Amstelveen                     0.42517
## districtGemeente Amsterdam                      0.22183
## districtGemeente Diemen                         0.01316 *
## districtGemeente Landsmeer                      0.81413
## districtGemeente Ouder-Amstel                   0.08194 .
## districtGemeente Waterland                      0.72334
## districtGemeente Zaanstad                            NA
## districtGreater London                               NA
## districtLisbon                                  0.12167
## districtLoures                                  0.00017 ***
## districtNomarchia Athinas                            NA
## districtOdivelas                                0.22176
## districtOeiras                                 1.66e-09 ***
## districtParis                                        NA
## districtPolitischer Bezirk Ganserndorf          0.00186 **
## districtPolitischer Bezirk Korneuburg           0.99613
## districtPolitischer Bezirk Modling             4.50e-09 ***
## districtPolitischer Bezirk Wien-Umgebung        0.10550
## districtProvincia de Barcelona                       NA
## districtRome                                         NA
## districtWien Stadt                                   NA
## stateBerlin                                     0.83211
## stateBrandenburg                                     NA
## stateBudapest                                   < 2e-16 ***
## stateCatalonia                                       NA
## stateEngland                                         NA
```

```
## stateIle-de-France                                    NA
## stateLatium                                           NA
## stateLisbon                                           NA
## stateLower Austria                                    NA
## stateNorth Holland                                    NA
## statePest                                             NA
## stateRome                                             NA
## stateSetubal                                          NA
## stateVienna                                           NA
## country_codeDE                                        NA
## country_codeES                                        NA
## country_codeFR                                        NA
## country_codeGB                                        NA
## country_codeGR                                        NA
## country_codeHU                                        NA
## country_codeIT                                        NA
## country_codeNL                                        NA
## country_codePT                                        NA
## country_codeVA                                        NA
## country_nameFrance                                    NA
## country_nameGermany                                   NA
## country_nameGreece                                    NA
## country_nameHoly See (Vatican City State)             NA
## country_nameHungary                                   NA
## country_nameItaly                                     NA
## country_nameNetherlands                               NA
## country_namePortugal                                  NA
## country_nameSpain                                     NA
## country_nameUnited Kingdom                            NA
## Crime_Index                                      0.56090
## Safety_Index                                     0.70134
## Monthly_Average_Net_salary                            NA
## Meal_at_Inexpensive_Restaurant                        NA
## Taxi_price_per_Km                                     NA
## Monthly_Basic_Utilities                          0.89243
## Monthly_Rent_One_Bedroom_CC                           NA
## Monthly_Rent_One_Bedroom_OCC                          NA
## Monthly_Rent_Three_Bedroom_CC                         NA
## Monthly_Rent_Three_Bedroom_OCC                        NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3214 on 36143 degrees of freedom
## Multiple R-squared:  0.837,  Adjusted R-squared:  0.8367
## F-statistic:  3569 on 52 and 36143 DF,  p-value: < 2.2e-16

#model comparison
AIC(lm_model, step_model)
```

```
##            df      AIC
## lm_model   55 16272.69
## step_model 54 20610.71
```

```
BIC(lm_model, step_model)
```

```
##            df      BIC
## lm_model   55 16740.01
## step_model 54 21069.53
```

Regularisation

```
x_train <- model.matrix(log_price ~ . - price_total, train)[,-1]

y_train <- train$log_price

x_test <- model.matrix(log_price ~ . - price_total, test)[,-1]
y_test <- test$log_price
```

#Ridge Regression

```
ridge_model <- cv.glmnet(x_train, y_train, alpha = 0)

ridge_pred <- predict(ridge_model, s = "lambda.min", newx = x_test)

cat("Ridge RMSE:", rmse(y_test, ridge_pred), "\n")
```

```
## Ridge RMSE: 0.3562661
```

#Lasso

```
lasso_model <- cv.glmnet(x_train, y_train, alpha = 1)

lasso_pred <- predict(lasso_model, s = "lambda.min", newx = x_test)

cat("LASSO RMSE:", rmse(y_test, lasso_pred), "\n")
```

```
## LASSO RMSE: 0.3552045
```

#Elastic Net

```
elastic_model <- cv.glmnet(x_train, y_train, alpha = 0.5)

elastic_pred <- predict(elastic_model, s = "lambda.min", newx = x_test)

cat("Elastic Net RMSE:", rmse(y_test, elastic_pred), "\n")
```

```
## Elastic Net RMSE: 0.3552133
```

PHASE 7: KNN REGRESSION

#Normalize Data

```r
preProc <- preProcess(train[, -c(which(names(train) %in%
c("price","log_price")))], method = c("center", "scale"))

train_scaled <- predict(preProc, train)
test_scaled  <- predict(preProc, test)
```

#Apply KNN

```r
knn_model <- knnreg(log_price ~ . - price_total, data = train_scaled, k = 5)

knn_pred <- predict(knn_model, newdata = test_scaled)

cat("KNN RMSE:", rmse(test$log_price, knn_pred), "\n")
```

```
## KNN RMSE: 0.3365121
```

PHASE 8: GLM (UNIT 5)

```r
#Fit GLM
glm_model <- glm(log_price ~ . - price_total, data = train, family =
gaussian(link = "identity"))

summary(glm_model)
```

```
##
## Call:
## glm(formula = log_price ~ . - price_total, family = gaussian(link =
"identity"),
##     data = train)
##
## Coefficients: (50 not defined because of singularities)
##                                          Estimate Std. Error t value
## (Intercept)                             6.432e+10  1.206e+11   0.533
## room_typePrivate room                  -4.126e-01  6.277e-03 -65.727
## room_typeShared room                   -8.206e-01  2.394e-02 -34.279
## is_shared_room                                 NA         NA      NA
## is_private_room                                NA         NA      NA
## max_guests                              9.083e-02  2.172e-03  41.824
## is_superhost                           -2.460e-02  5.499e-03  -4.475
## is_multi_listing                        4.956e-02  4.732e-03  10.473
## is_business_listing                     1.201e-01  4.786e-03  25.090
## cleanliness_score                       3.897e-02  2.773e-03  14.055
## guest_satisfaction_score                8.516e-04  3.020e-04   2.819
## num_bedrooms                            1.493e-01  3.667e-03  40.714
## distance_city_center                   -5.492e-02  1.712e-03 -32.084
## distance_metro                         -1.789e-01  3.531e-03 -50.661
## attraction_index                        9.477e-05  2.557e-05   3.706
## attraction_index_norm                   7.571e-03  5.655e-04  13.389
## restaurant_index                        9.807e-05  1.114e-05   8.802
## restaurant_index_norm                   4.959e-04  2.751e-04   1.803
## longitude                              -8.975e-01  5.279e-02 -17.001
```

```
## latitude                                          2.877e-02  9.350e-02   0.308
## cityAthens                                        2.145e+10  3.864e+10   0.555
## cityBarcelona                                     2.457e+10  2.765e+10   0.889
## cityBerlin                                       -8.104e+09  6.181e+10  -0.131
## cityBudapest                                      2.305e+10  2.596e+10   0.888
## cityLisbon                                        2.217e+10  2.697e+10   0.822
## cityLondon                                       -2.505e+09  7.941e+10  -0.032
## cityParis                                         1.359e+10  5.749e+10   0.236
## cityRome                                          1.861e+10  2.457e+10   0.757
## cityVienna                                       -3.913e+09  1.400e+10  -0.279
## day_typeweekend                                   2.539e-02  3.684e-03   6.892
## districtBerlin                                           NA         NA      NA
## districtBudapest                                         NA         NA      NA
## districtCitta metropolitana di Roma Capitale      1.397e-02  1.086e-02   1.286
## districtDepartement de Seine-Saint-Denis         -1.235e-01  1.694e-02  -7.292
## districtDepartement des Hauts-de-Seine            1.210e-01  1.305e-02   9.269
## districtDepartement du Val-de-Marne              -1.019e-02  2.129e-02  -0.478
## districtGemeente Aalsmeer                        -9.279e-02  2.538e-01  -0.366
## districtGemeente Amstelveen                       5.710e-04  7.551e-02   0.008
## districtGemeente Amsterdam                        2.707e-01  6.286e-02   4.306
## districtGemeente Diemen                           2.288e-01  7.291e-02   3.139
## districtGemeente Landsmeer                        1.246e-02  7.905e-02   0.158
## districtGemeente Ouder-Amstel                     2.356e-01  7.158e-02   3.292
## districtGemeente Waterland                       -2.744e-03  1.677e-01  -0.016
## districtGemeente Zaanstad                                NA         NA      NA
## districtGreater London                                   NA         NA      NA
## districtLisbon                                   -1.002e-01  2.070e-02  -4.842
## districtLoures                                    5.330e-02  3.462e-02   1.540
## districtNomarchia Athinas                                NA         NA      NA
## districtOdivelas                                  3.189e-02  4.585e-02   0.695
## districtOeiras                                    4.894e-02  3.844e-02   1.273
## districtParis                                            NA         NA      NA
## districtPolitischer Bezirk Ganserndorf           -9.176e-02  6.339e-02  -1.448
## districtPolitischer Bezirk Korneuburg             6.005e-02  1.011e-01   0.594
## districtPolitischer Bezirk Modling               -1.444e-01  3.673e-02  -3.930
## districtPolitischer Bezirk Wien-Umgebung          1.416e-02  4.109e-02   0.345
## districtProvincia de Barcelona                           NA         NA      NA
## districtRome                                             NA         NA      NA
## districtWien Stadt                                       NA         NA      NA
## stateBerlin                                       2.769e-01  1.567e-01   1.767
## stateBrandenburg                                         NA         NA      NA
## stateBudapest                                    -4.808e-01  8.684e-02  -5.536
## stateCatalonia                                           NA         NA      NA
## stateEngland                                             NA         NA      NA
## stateIle-de-France                                       NA         NA      NA
## stateLatium                                              NA         NA      NA
## stateLisbon                                              NA         NA      NA
## stateLower Austria                                       NA         NA      NA
## stateNorth Holland                                       NA         NA      NA
## statePest                                                NA         NA      NA
```

```
## stateRome                                          NA          NA      NA
## stateSetubal                                       NA          NA      NA
## stateVienna                                        NA          NA      NA
## country_codeDE                                     NA          NA      NA
## country_codeES                                     NA          NA      NA
## country_codeFR                                     NA          NA      NA
## country_codeGB                                     NA          NA      NA
## country_codeGR                                     NA          NA      NA
## country_codeHU                                     NA          NA      NA
## country_codeIT                                     NA          NA      NA
## country_codeNL                                     NA          NA      NA
## country_codePT                                     NA          NA      NA
## country_codeVA                                     NA          NA      NA
## country_nameFrance                                 NA          NA      NA
## country_nameGermany                                NA          NA      NA
## country_nameGreece                                 NA          NA      NA
## country_nameHoly See (Vatican City State)          NA          NA      NA
## country_nameHungary                                NA          NA      NA
## country_nameItaly                                  NA          NA      NA
## country_nameNetherlands                            NA          NA      NA
## country_namePortugal                               NA          NA      NA
## country_nameSpain                                  NA          NA      NA
## country_nameUnited Kingdom                         NA          NA      NA
## Crime_Index                                -1.292e+09  2.022e+09  -0.639
## Safety_Index                               -9.899e+08  9.817e+08  -1.008
## Monthly_Average_Net_salary                         NA          NA      NA
## Meal_at_Inexpensive_Restaurant                     NA          NA      NA
## Taxi_price_per_Km                                  NA          NA      NA
## Monthly_Basic_Utilities                     1.619e+08  3.094e+08   0.523
## Monthly_Rent_One_Bedroom_CC                        NA          NA      NA
## Monthly_Rent_One_Bedroom_OCC                       NA          NA      NA
## Monthly_Rent_Three_Bedroom_CC                      NA          NA      NA
## Monthly_Rent_Three_Bedroom_OCC                     NA          NA      NA
## proximity_index                             1.159e+00  1.491e-02  77.704
##                                            Pr(>|t|)
## (Intercept)                                0.593696
## room_typePrivate room                       < 2e-16 ***
## room_typeShared room                        < 2e-16 ***
## is_shared_room                                   NA
## is_private_room                                  NA
## max_guests                                  < 2e-16 ***
## is_superhost                               7.68e-06 ***
## is_multi_listing                            < 2e-16 ***
## is_business_listing                         < 2e-16 ***
## cleanliness_score                           < 2e-16 ***
## guest_satisfaction_score                   0.004815 **
## num_bedrooms                                < 2e-16 ***
## distance_city_center                        < 2e-16 ***
## distance_metro                              < 2e-16 ***
## attraction_index                           0.000211 ***
```

```
## attraction_index_norm                            < 2e-16 ***
## restaurant_index                                 < 2e-16 ***
## restaurant_index_norm                           0.071417 .
## longitude                                        < 2e-16 ***
## latitude                                        0.758281
## cityAthens                                      0.578826
## cityBarcelona                                   0.374204
## cityBerlin                                      0.895682
## cityBudapest                                    0.374455
## cityLisbon                                      0.410921
## cityLondon                                      0.974839
## cityParis                                       0.813104
## cityRome                                        0.448933
## cityVienna                                      0.779890
## day_typeweekend                                 5.59e-12 ***
## districtBerlin                                        NA
## districtBudapest                                      NA
## districtCitta metropolitana di Roma Capitale    0.198544
## districtDepartement de Seine-Saint-Denis        3.12e-13 ***
## districtDepartement des Hauts-de-Seine           < 2e-16 ***
## districtDepartement du Val-de-Marne             0.632311
## districtGemeente Aalsmeer                       0.714671
## districtGemeente Amstelveen                     0.993967
## districtGemeente Amsterdam                      1.67e-05 ***
## districtGemeente Diemen                         0.001698 **
## districtGemeente Landsmeer                      0.874755
## districtGemeente Ouder-Amstel                   0.000996 ***
## districtGemeente Waterland                      0.986940
## districtGemeente Zaanstad                             NA
## districtGreater London                                NA
## districtLisbon                                  1.29e-06 ***
## districtLoures                                  0.123602
## districtNomarchia Athinas                             NA
## districtOdivelas                                0.486773
## districtOeiras                                  0.202985
## districtParis                                         NA
## districtPolitischer Bezirk Ganserndorf          0.147723
## districtPolitischer Bezirk Korneuburg           0.552627
## districtPolitischer Bezirk Modling              8.50e-05 ***
## districtPolitischer Bezirk Wien-Umgebung        0.730449
## districtProvincia de Barcelona                        NA
## districtRome                                          NA
## districtWien Stadt                                    NA
## stateBerlin                                     0.077174 .
## stateBrandenburg                                      NA
## stateBudapest                                   3.11e-08 ***
## stateCatalonia                                        NA
## stateEngland                                          NA
## stateIle-de-France                                    NA
## stateLatium                                           NA
```

```
## stateLisbon                                          NA
## stateLower Austria                                   NA
## stateNorth Holland                                   NA
## statePest                                            NA
## stateRome                                            NA
## stateSetubal                                         NA
## stateVienna                                          NA
## country_codeDE                                       NA
## country_codeES                                       NA
## country_codeFR                                       NA
## country_codeGB                                       NA
## country_codeGR                                       NA
## country_codeHU                                       NA
## country_codeIT                                       NA
## country_codeNL                                       NA
## country_codePT                                       NA
## country_codeVA                                       NA
## country_nameFrance                                   NA
## country_nameGermany                                  NA
## country_nameGreece                                   NA
## country_nameHoly See (Vatican City State)            NA
## country_nameHungary                                  NA
## country_nameItaly                                    NA
## country_nameNetherlands                              NA
## country_namePortugal                                 NA
## country_nameSpain                                    NA
## country_nameUnited Kingdom                           NA
## Crime_Index                                    0.522931
## Safety_Index                                   0.313299
## Monthly_Average_Net_salary                           NA
## Meal_at_Inexpensive_Restaurant                       NA
## Taxi_price_per_Km                                    NA
## Monthly_Basic_Utilities                        0.600701
## Monthly_Rent_One_Bedroom_CC                          NA
## Monthly_Rent_One_Bedroom_OCC                         NA
## Monthly_Rent_Three_Bedroom_CC                        NA
## Monthly_Rent_Three_Bedroom_OCC                       NA
## proximity_index                                 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for gaussian family taken to be 0.1211501)
##
##     Null deviance: 22905.6  on 36195  degrees of freedom
## Residual deviance:  4378.7  on 36143  degrees of freedom
## AIC: 26375
##
## Number of Fisher Scoring iterations: 2
```

#Model Comparison

```r
results <- data.frame(
  Model = c("Linear", "Stepwise", "Ridge", "LASSO", "Elastic", "KNN"),
  RMSE = c(
    rmse(test$log_price, pred_lm),
    rmse(test$log_price, predict(step_model, test)),
    rmse(y_test, ridge_pred),
    rmse(y_test, lasso_pred),
    rmse(y_test, elastic_pred),
    rmse(test$log_price, knn_pred)
  )
)

print(results[order(results$RMSE), ])
```

```
##      Model      RMSE
## 1   Linear 0.2975413
## 2 Stepwise 0.3148189
## 6      KNN 0.3365121
## 4    LASSO 0.3552045
## 5  Elastic 0.3552133
## 3    Ridge 0.3562661
```

```r
#final model

best_model <- step_model  # or whichever has lowest RMSE

final_pred <- predict(best_model, newdata = test)
```